



41ST IN-SERVICE TRAINING COURSE FOR JUNIOR ENGINEERS · BS-17 TO BS-18

AI Is Already Replacing Tasks, Not Humans

Practical AI agents for the modern government engineer — from repetitive paperwork to AI-assisted workflows to accountable, verified agentic engineering.

Punjab Engineering Academy, Lahore · 27 August 2026 · C&W · Public Health Engineering · GDA · Punjab C&W

YOUR INSTRUCTOR TODAY

Engr. Muhammad Ansar



LAHORE, PAKISTAN

Head of AI · Nexum Global Solutions
PhD Researcher, Energy Engineering · University of the Punjab, Lahore

An electrical engineer working where power systems, renewable energy, industrial automation and applied AI meet. My practice builds the kind of system you will see running today — document intelligence, data pipelines, computer vision and LLM agents — for industry and for government departments, alongside energy and automation consulting and technical training for engineering teams.

Electrical & Power Systems

Solar PV · Net Metering · Storage

PLC · SCADA · Automation

Machine Learning & Computer Vision

AI Product Development

 EMAIL
engr.mianansar24@gmail.com

 CALL / WHATSAPP
0348 3588143

 LINKEDIN
Muhammad Ansar

 PORTFOLIO
Full profile & services

THREE HOURS, IN THREE PARTS

How today runs

0:00 – 0:55

Part 1 – The Shift

From chatbot to accountable AI agent, demonstrated live on a real engineering case file.

0:55 – 2:10

Part 2 – Build

A working Government Engineering AI Copilot — document intelligence, BOQ checks, progress monitoring, ML delay risk. Break included.

2:10 – 3:00

Part 3 – Judgment

Human-in-the-loop, risk, and a practical challenge: design an AI agent for your own department.

PART 1 / 3

Where Engineers Actually Spend Their Day

Not "what is artificial intelligence" — a look at the
actual workday, task by task.

IN THIS ROOM

Four departments, one common denominator

C&W

Roads & Buildings

Site inspections, BOQs, measurement, contractor documentation, progress monitoring, tender specs.

PHE

Public Health Engineering

Water supply, sewerage & drainage, pumping systems, field inspections, asset maintenance.

GDA

Development Authority

Development approvals, site inspection, GIS & spatial data, compliance monitoring.

PUNJAB C&W

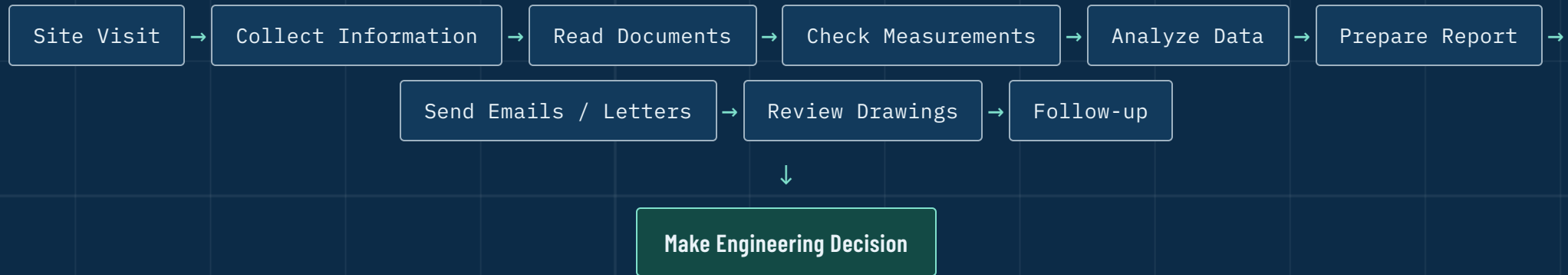
Project Execution

Measurement, contractor management, quality assurance, progress reporting, maintenance.

Documents + Data + Field Information + Engineering Judgment — every workflow in this room is built from the same four ingredients. That is the foundation for everything today.

START HERE, NOT WITH "WHAT IS AI"

Where does an engineer actually spend the day?

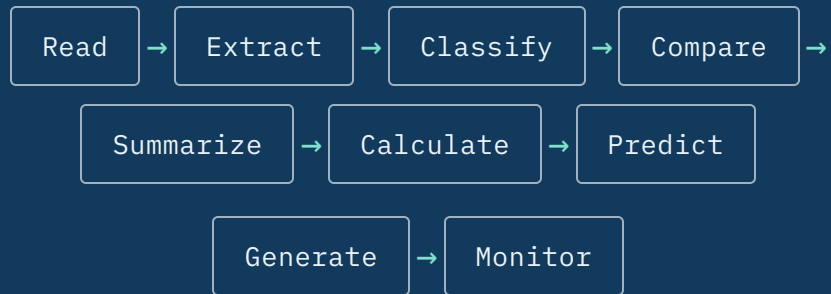


"Which of these steps require an engineer — and which simply consume an engineer's time?"

That question is the entire workshop. Everything before the last box is information processing. Only the last box is engineering.

Tasks vs. humans

WHAT AI IS GOOD AT



WHAT ENGINEERS ARE ESSENTIAL FOR



**AI will not replace the
engineer who makes
the decision. It will
replace the repetitive
work that surrounds it.**

Let AI process information. Let engineers make accountable
decisions.

AI can already do this

CASE FILE

Rehabilitation of Link Road, Chowk Azam – Kot Sultan (12.4 km)

District Layyah · Punjab C&W Department

Project brief

Approved BOQ

Contractor progress report

Site inspection notes

Measurement / payment record

1 · Extract – "Pull every project parameter from this document."

2 · Compare – "Compare planned progress against reported progress."

3 · Detect – "Flag inconsistencies, missing information, and risk."

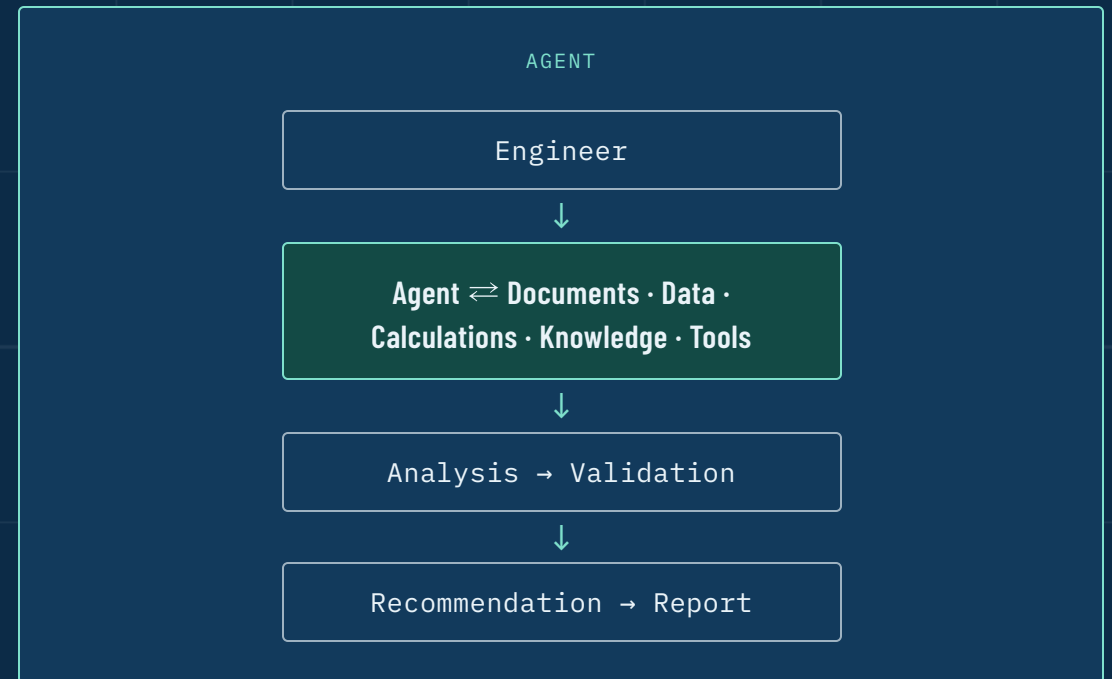
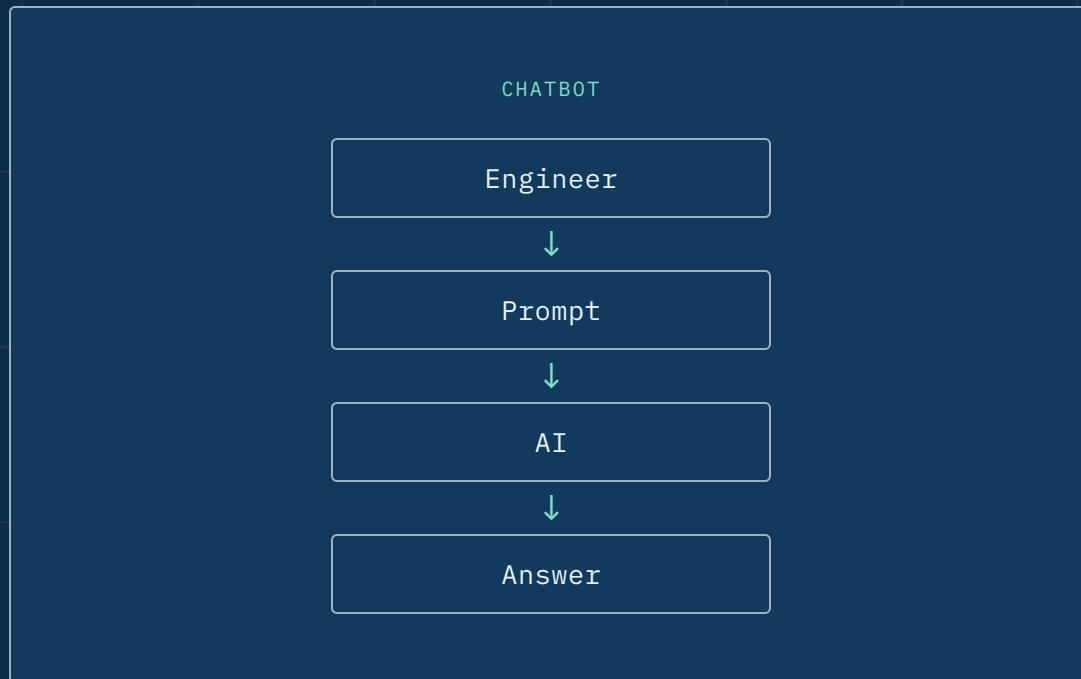
4 · Generate – "Produce a site inspection checklist."

5 · Report – "Draft a one-page progress report for the SDO."

//

**I have not replaced the
SDO. I have reduced
the SDO's information-
processing workload.**

From chatbot to agent



A chatbot answers questions. An agent is built to accomplish a task — and hand back something an engineer can verify.

PART 2 / 3

Build: The Government Engineering AI Copilot

One architecture. Every department. Live, on this
projector.

The Government Engineering AI Copilot

One Streamlit application, six modules, loaded live with the exact Road Rehabilitation case file from Part 1 — nothing pre-recorded.

- Python
- Streamlit
- Groq API
- Pandas
- scikit-learn

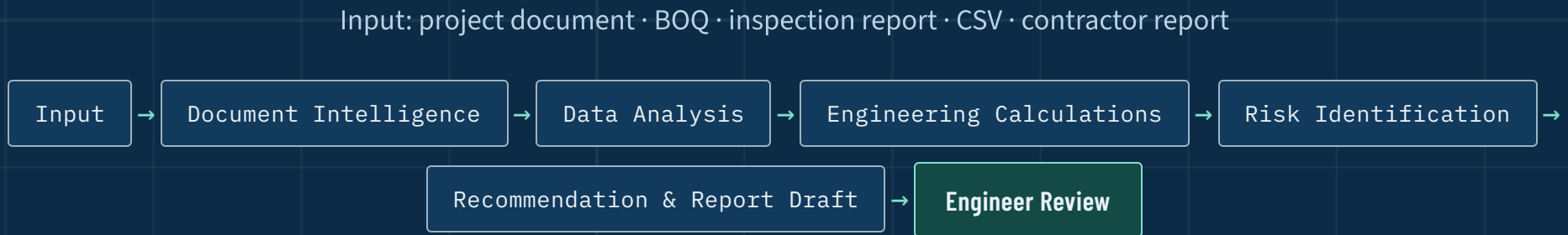
GROUND RULE

Every output the copilot produces is a draft for review. Nothing here signs, approves, or certifies anything — that stays with you.



FIG. 01 – LIVE APPLICATION, DOCUMENT ANALYZER

How the copilot processes a project



What the copilot actually does

01

Document Analyzer

Extracts scope, contractor, timeline, cost and milestones from a project document.

02

BOQ Assistant

Structures BOQ items, flags unusual rates — verified against the approved BOQ.

03

Progress Monitor

Planned vs. actual progress, variance, and a plain-language risk flag.

04

Inspection Assistant

Field notes in, structured checklist and action items out.

05

Report Generator

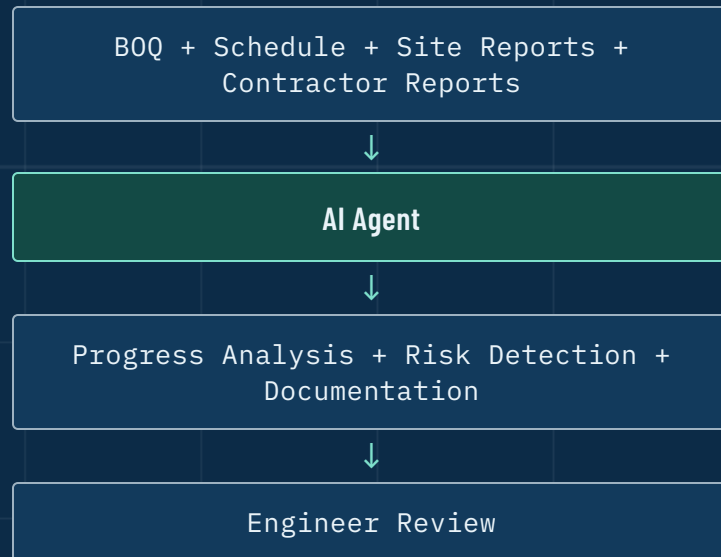
Field notes and data become a one-page draft report for the supervising engineer.

06

Delay Risk (ML)

A small model trained on past projects scores schedule risk on the current one.

AI Construction Project Agent



- Progress reporting, drafted in minutes not hours
- BOQ analysis and quantity cross-checks
- Delay identification against the approved schedule
- Contractor report analysis
- Quality-control documentation

AI Water & Public Health Engineering Agent

Input

- Pump run-hours and flow data
- Water consumption records
- Maintenance and inspection logs

Assists with

- Pump performance analysis and maintenance scheduling
- Water-demand analysis and anomaly detection
- Asset documentation and reporting

LIVE PROMPT

"Here is a month of pump and flow data. Find the abnormal readings."

AI Urban Development Assistant

- Development project documentation and site inspection
- Compliance checklist generation
- Infrastructure monitoring and GIS-related interpretation

NEXT HORIZON – NOT BUILT TODAY

Satellite / Drone
/ Site Images



Computer Vision



Detect Changes



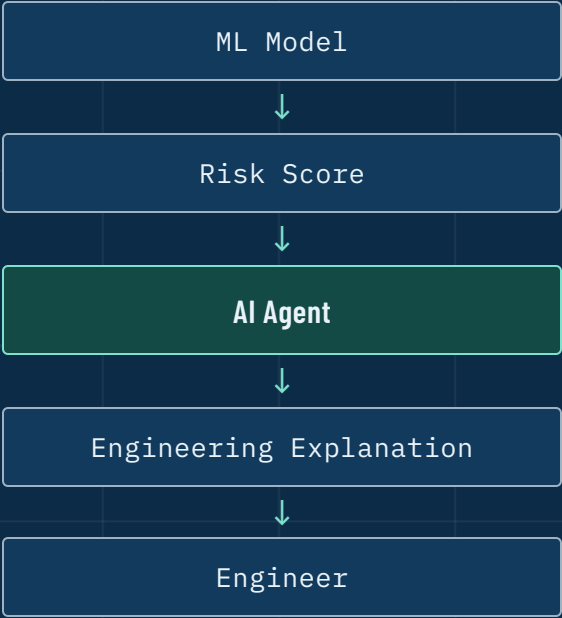
AI Agent



Engineering
Assessment

Predicting delay risk from project history

PROJECT	DURATION	PLANNED %	ACTUAL %	DELAY
Link Road — Layyah	10 mo	72%	58%	?
Farm-to-Market Rd A	12 mo	90%	75%	3 mo
Sewerage Scheme B	18 mo	80%	60%	5 mo
Bridge Rehab C	10 mo	95%	92%	1 mo



The model scores the risk. The agent explains it in engineering terms. The engineer decides what to do about it.

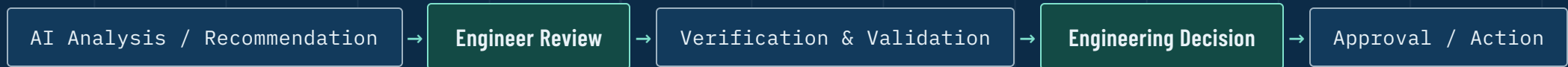
PART 3 / 3

Judgment, Risk, and Practice

What to trust, what to verify, and what you'll build for
your own department.

THE MOST IMPORTANT SLIDE TODAY

Human-in-the-loop, by design



AI can recommend. The engineer remains accountable – every time.

Six risks to carry back to your office

Hallucination

AI can produce technically plausible information that is simply wrong.

Confidentiality

Government documents can carry sensitive information — mind what you paste in.

Incorrect calculations

Never trust an LLM's arithmetic on a critical figure without checking it.

Outdated standards

AI won't know your latest departmental SOPs unless you give them to it.

Bias

Recommendations can inherit bias from the data or documents they're built on.

Accountability

The AI does not hold a BS-17 post, and it does not answer an audit query.

Who signs the engineering document? Not the AI.

Find a task to replace, not a job



GROUP A – C&W

How can AI reduce the time required for site progress reporting?

GROUP B – PHE

How can AI monitor water and pumping infrastructure?

GROUP C – GDA

How can AI assist development and project inspection?

GROUP D – CROSS-DISCIPLINARY

How can AI identify project delays before they become serious?

WHERE THIS IS HEADED

Three stages of engineering practice

STAGE 1

Manual Engineering

Engineer + documents + Excel + manual analysis.

STAGE 2

AI-Assisted Engineering

+ AI tools, automation, data analytics.

STAGE 3 – TODAY'S BUILD

Agentic Engineering

Engineer → Engineering AI Agent → Data + Documents
+ Models + Tools → Continuous Analysis →
Recommendation → Engineer.

**The engineer of the
future will not compete
against AI. The
engineer of the future
will know how to
command, verify, and
engineer AI.**

BEFORE YOU GO

Want one of these built for your department?

Today's copilot was fitted to one case file from Layyah. The same architecture can be fitted to **your** office — your BOQ format, your inspection proforma, your monthly progress return. Easiest way to start: bring one real workflow — one recurring report, one document you read every week — and we scope it together.

BUILD

Departmental AI Copilot

Document intelligence, BOQ and measurement checks, progress monitoring and report drafting — trained on your own formats and case files.

DATA

Automation & Analytics

Reporting pipelines, dashboards, GIS and spatial data, and ML models for delay, cost and quality risk across a portfolio of schemes.

ENGINEERING

Energy & Automation Consulting

Electrical design and load studies, solar PV with net metering and storage, microgrids, PLC/SCADA and industrial automation.

TRAIN

Workshops for Your Wing

On-site or online AI training for engineering staff, built around the actual documents and decisions your department handles.

EMAIL

 engr.mianansar24@gmail.com

CALL / WHATSAPP

 0348 3588143

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 Muhammad Ansar

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Thank you

Engr. Muhammad Ansar · Sample case files and the Copilot app are available on request



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engr.mianansar24@gmail.com



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